

FROM DATA TO INSIGHTS:

Working with DestinE Data Lake Services

PART 1

Welcome to DestinE

Ben Loveday (Innoflair / EUMETSAT)
EUMETSAT Conference 2026





- Our Agenda
 - Part 1: Welcome to DestinE
 - Part 2: Introduction to the EUMETSAT Data Lake: holdings, services and onboarding (**demonstration**)
 - Part 3: Data Discovery with the HDA (**demonstration**)
 - Part 4: Data Processing with the Data Lake Edge Services (**demonstration**)
 - Part 5: New tools and services: DEFAIR, DEDL Cube & MLOPS
 - Part 6: Conclusion and Q&A



- What are our goals today?
 - Provide an overview of DestinE and its **Data Lake**
 - Show how to **navigate, discover and onboard** to Data Lake services
 - Explore Harmonised Data Access (HDA) for efficient **data discovery, search and access**
 - Explore Edge Services for **processing and analysing data** close to where it is hosted
 - Demonstrate **practical workflows** that bring these capabilities together
 - Provide an opportunity for **questions, discussion and feedback**



How can I interact?

- Please feel free to ask questions in room!
- Alternatively (and preferably), we have an open Q&A on slido
- Feel free to speak to your trainers (below)



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Ben



Joana



Madalina



How can I interact?

- Let's start here —————→
- What do you already know about DestinE?
- What are you interested in learning more about?



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- A European Commission flagship initiative to create a highly accurate digital model of the Earth and support Europe's response to environmental challenges.

DESTINE AIMS TO CREATE A DIGITAL REPLICA OF THE EARTH SYSTEM TO HELP EUROPE UNDERSTAND HOW IT IS CHANGING, EXPLORE WHAT COULD HAPPEN NEXT, AND MAKE BETTER DECISIONS TO ADAPT

- **Observe:** bring together vast Earth-system observations and data
- **Simulate:** model the Earth system at unprecedented scale and detail
- **Explore:** interact with Digital Twins and test scenarios
- **Analyse:** combine data, models, AI and extreme-scale computing
- **Act:** turn Earth-system information into actionable knowledge for evidence-based decisions
- Currently in phase 3: Sustained operation and evolution (06.2026 – 06.2028), with Phase 4 under discussion



DestinE components

Three core components

Core Service Platform (DESP)

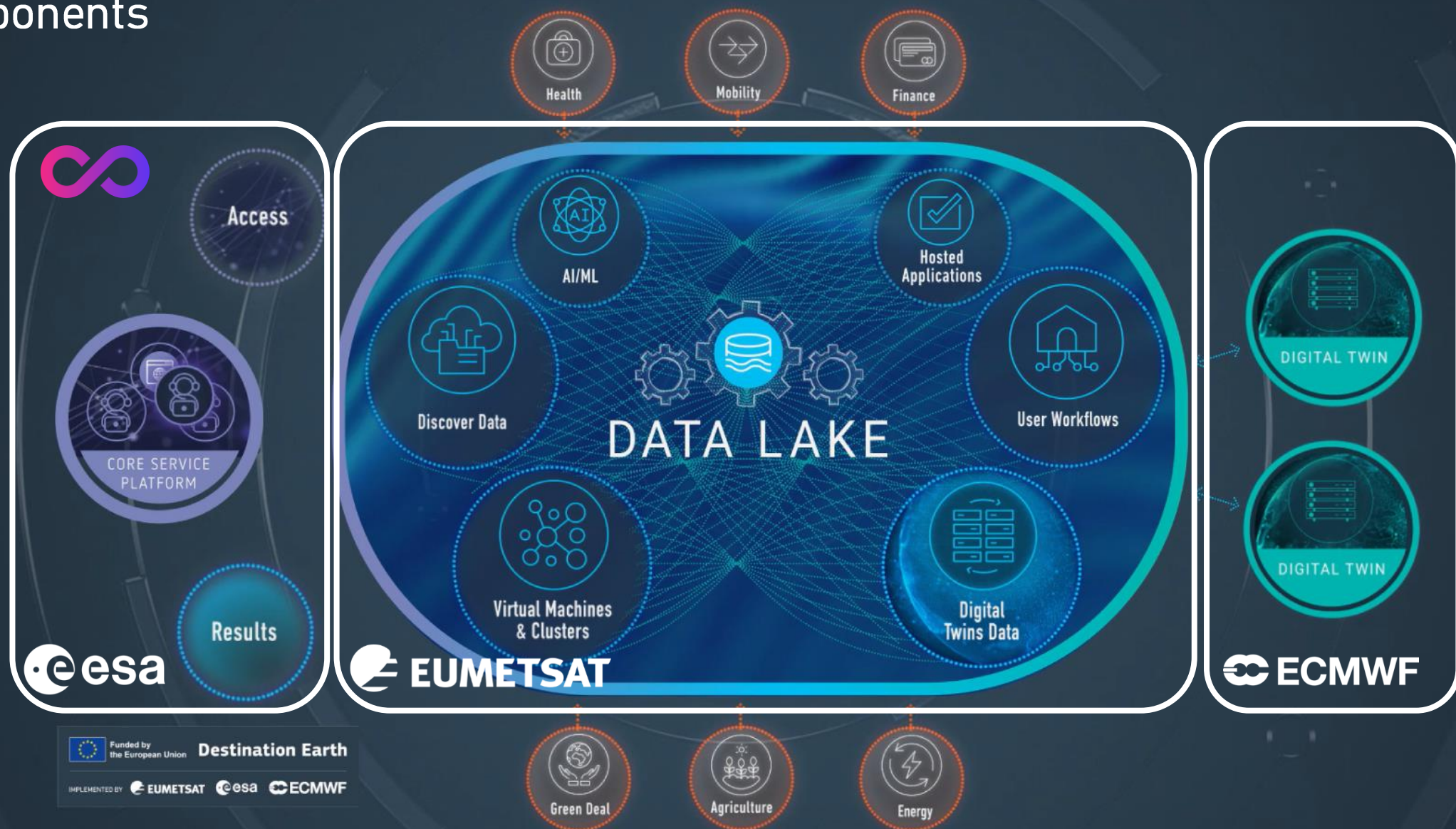
- Entry point for users
- Access to DestinE services, tools and applications

Data Lake (DEDL)

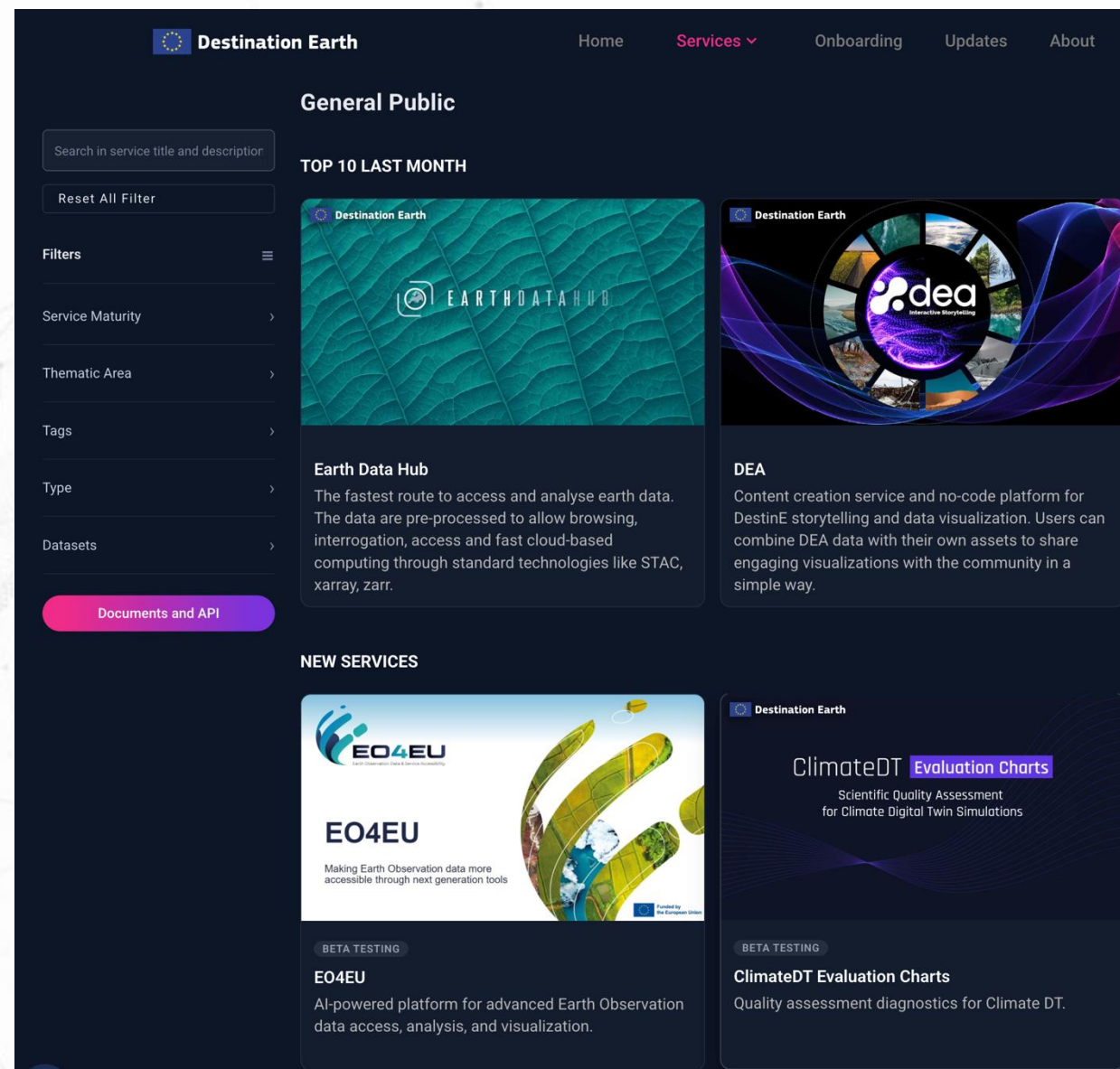
- Discovery, access and processing of Earth-system data
- Connects users to observations, models and simulations

Digital Twins (DT)

- High-resolution Earth-system simulations
- Explore change, extremes and future scenarios

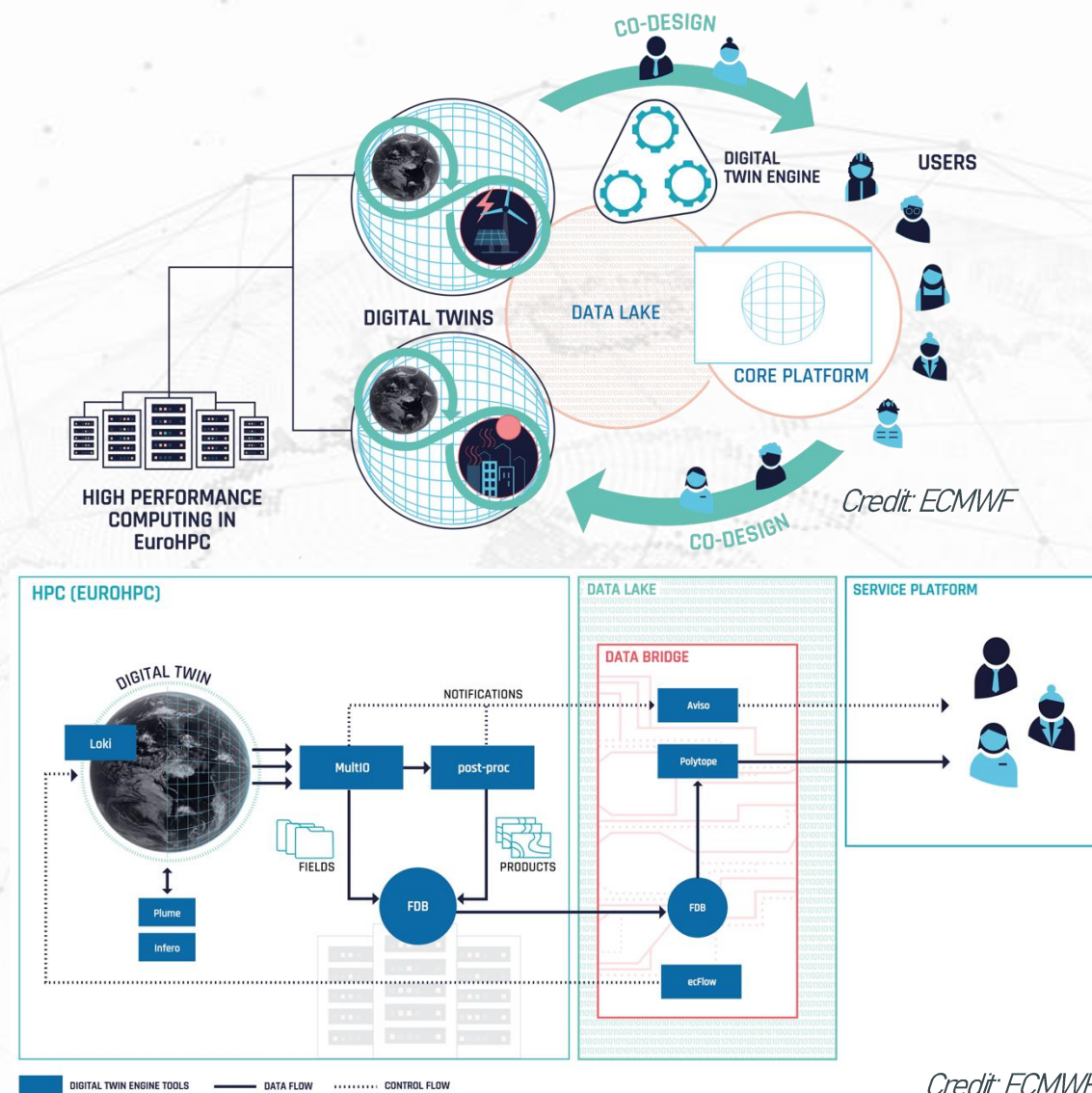


- The service platform can be found at <https://platform.destine.eu/>
- It presents a “marketplace” of available services; access, processing, application specific (this includes EUMETSAT Data Lake DEDL edge services)
- It exploits the DEDL holdings through various toolkits, though many services also extend to external resources.
- Services (including those prototyped on edge resources) can apply for registration on the DESP for operational use.
- Policy related services are an EC platform priority.



Digital Twins (DT) and engine (DTE)

- Digital twins are highly accurate replicas of the Earth system used to simulate, monitor and predict environmental change.
- More information on the DTs is available at <https://destine.ecmwf.int/digital-twins/>
- Currently two:
 - Climate: global, 5-10 km, multi-decadal, with "story lines"/scenarios
 - Extremes: 4.4/25 km, short-range.
 - More may come
- DT access (e.g. from DESP) is via the EUMETSAT Data Lake (DEDL)
- Adjacent EuroHPC runs DTs, but DestinE provides no HPC resources.



Credit: ECMWF



DestinE components

Three core components

Core Service Platform (DESP)

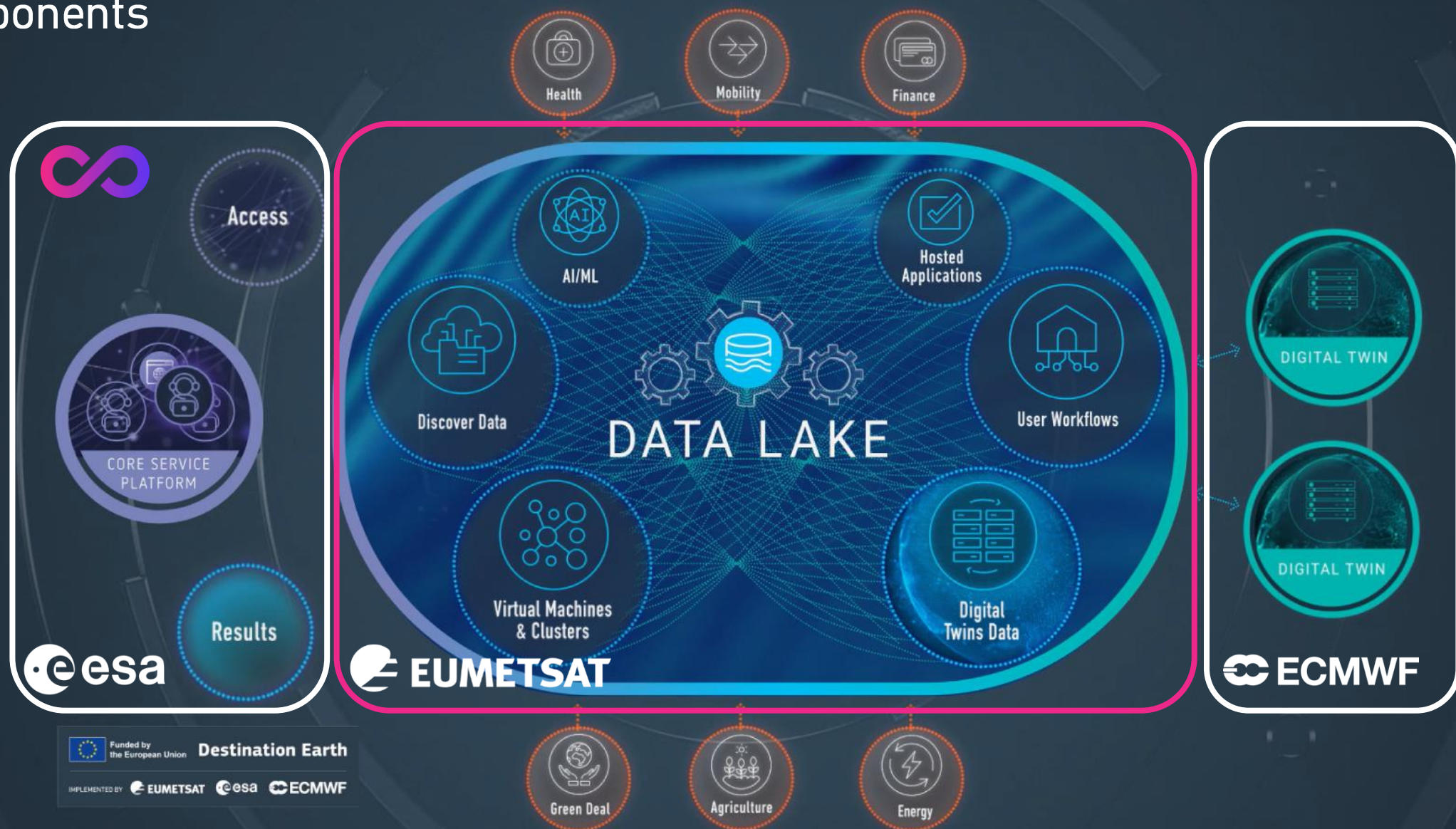
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Data Lake (DEDL)

- Discovery, access and processing of Earth-system data
- Connects users to observations, models and simulations

Digital Twins (DT)

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- Explore change, extremes and future scenarios



FROM DATA TO INSIGHTS:

Working with DestinE Data Lake Services

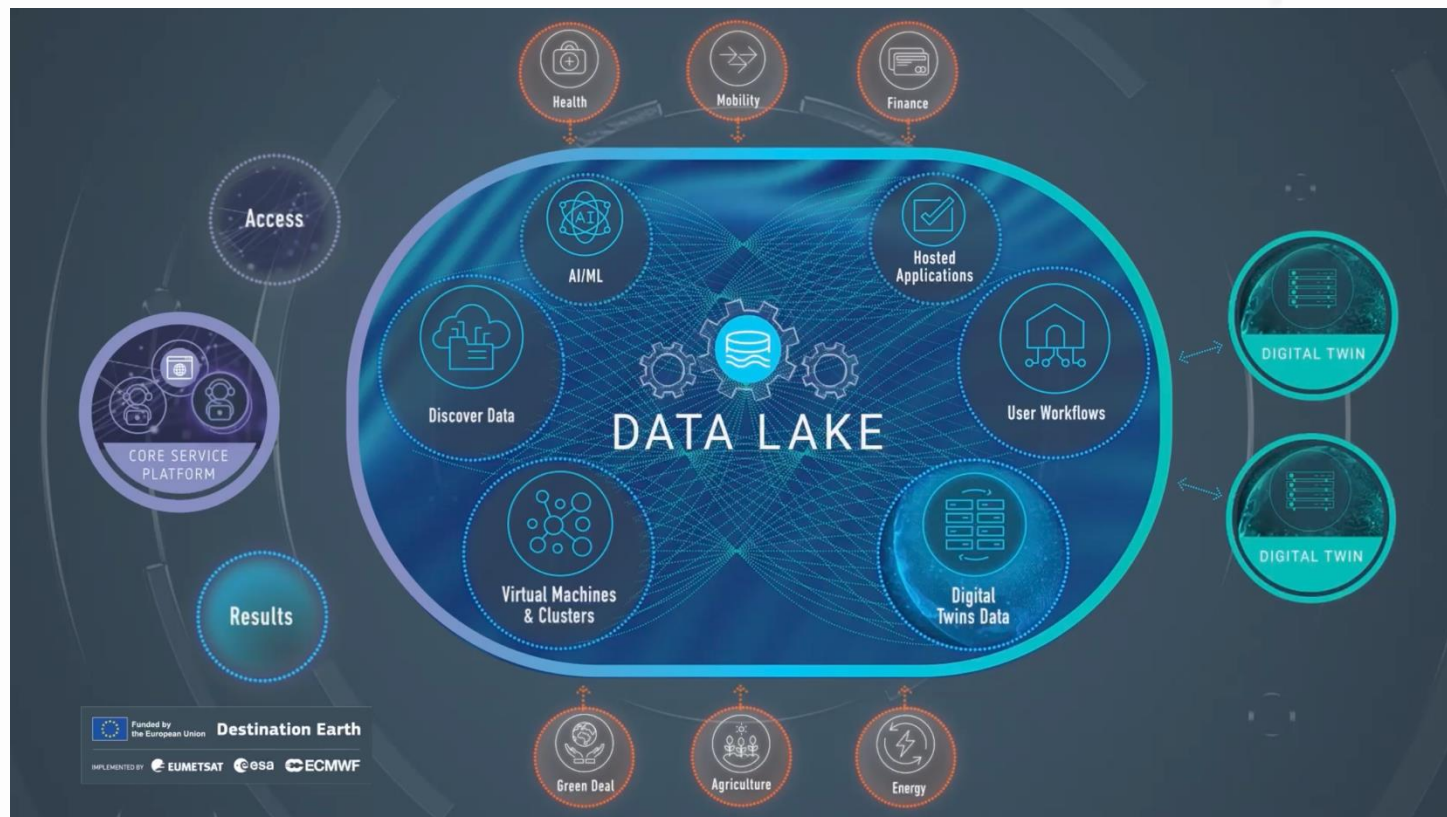
PART 2

The Data Lake: holdings, edge services and onboarding

Joana Brito (Innoflair / EUMETSAT)
EUMETSAT Conference 2026



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The DestinE Data Lake (DEDL) is essentially the data and data-processing infrastructure layer of DestinE.

It is not itself a Digital Twin or an end-user application,

its role is to make the huge volumes of Earth-system data required by DestinE available, discoverable, and processable at scale

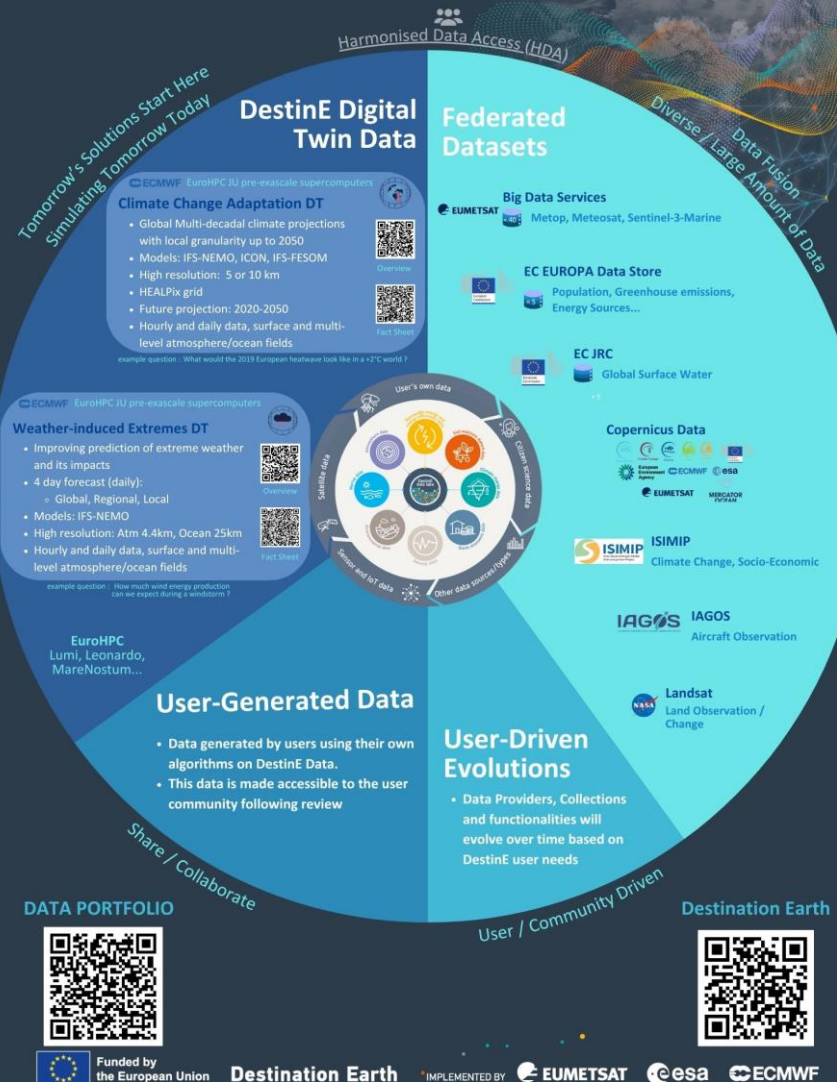


very diverse portfolio of data

earth observation
modelling
environmental
socio-economic

Atmosphere
Air Quality
Imagery
Demographic
Fire
Biomass/Vegetation
Ice
Soil
Land
Marine
Lightning
Climate/Weather
Water
Temperature
Land Cover
Wind

Destination Earth Data Portfolio



discoverable, accessible
and usable within the
same ecosystem



Funded by the European Union

Destination Earth

IMPLEMENTED BY





Federated Datasets

Federated datasets are datasets held by external data providers that are made available through DestinE, allowing users to discover and access them through the common DestinE Data Lake environment without necessarily having to physically move the data into the Data Lake

Big Data Services



EC EUROPA Data Store



Copernicus Data



European Environment Agency



EC JRC



IAGOS



ISIMIP



Landsat



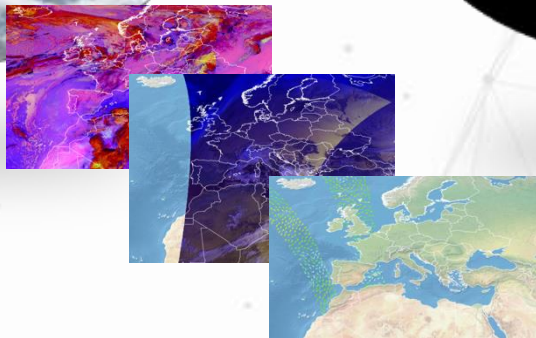


Federated Datasets

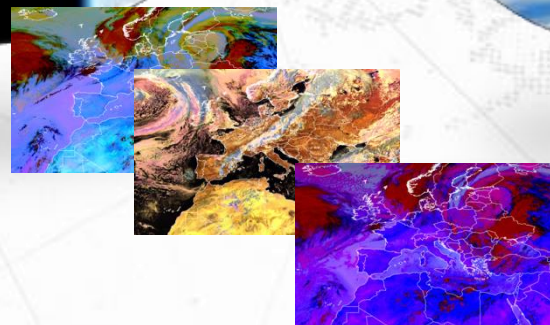
Big Data Services



Metop



Meteosat MSG



Sentinel-3





Federated Datasets

Big Data Services

Metop, Meteosat, Sentinel-3 Marine

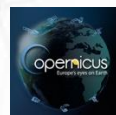


EC EUROPA Data Store

Population stats, Greenhouse Emissions, Energy Sources



Copernicus Data



European Environment Agency



EC JRC



IAGOS



ISIMIP



Lansat





Federated Datasets

Big Data Services

Metop, Meteosat, Sentinel-3 Marine

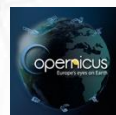


EC EUROPA Data Store

Population stats, Greenhouse Emissions, Energy Sources



Copernicus Data



European Environment Agency



EC JRC

Global Surface Water



IAGOS

Aircraft Observations



ISIMIP

Climate Change, Socio-Economic



Landsat

Land Observation/Change





DestinE Digital Twin Data



Climate Change Adaptation DT

- Global multi-decadal climate projections with local granularity up to 2050
- Models: IFS-NEMO, ICON, IFS-FESOM
- High resolution: 5 to 10 km
- Future Projection: 2020-2050
- Hourly and daily data, surface and multi-level atmosphere/ocean fields

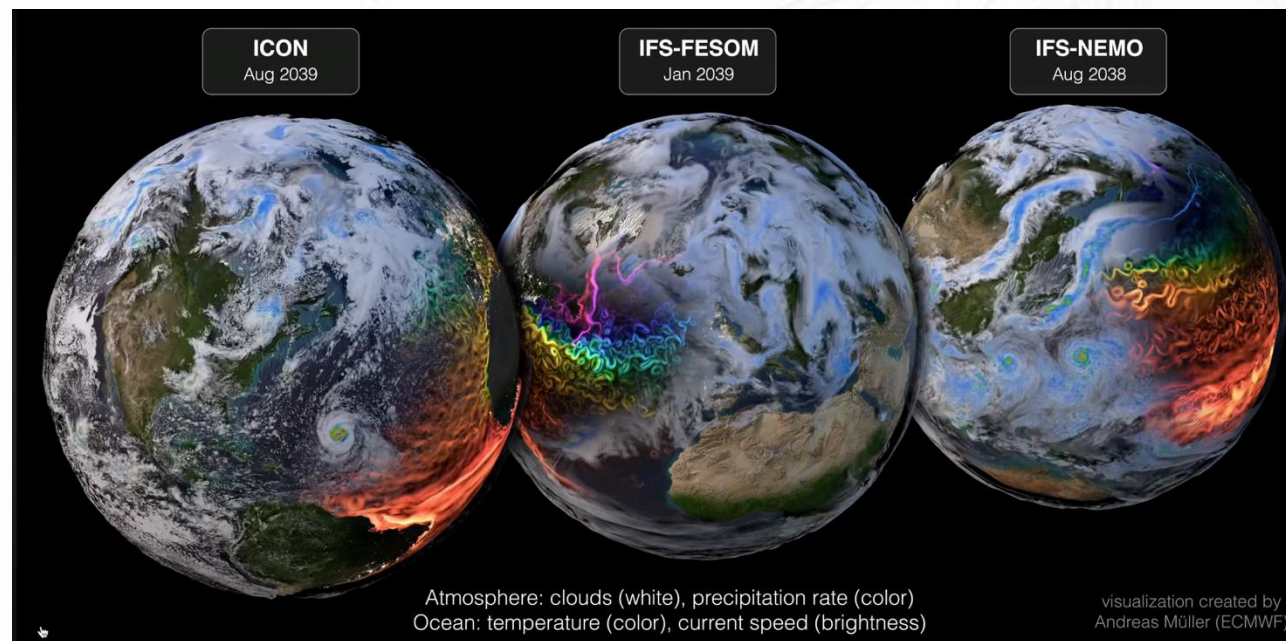
DestinE Digital Twin Data

Weather-induced Extremes DT

- Improving prediction of extreme weather and its impacts
- 4 day forecast (daily)
 - Global | Regional | Local
- High resolution: atm 4.4 km | ocean 25km
- Hourly and daily data, surface and multi-level atmosphere/ocean fields



DestinE Digital Twin Data

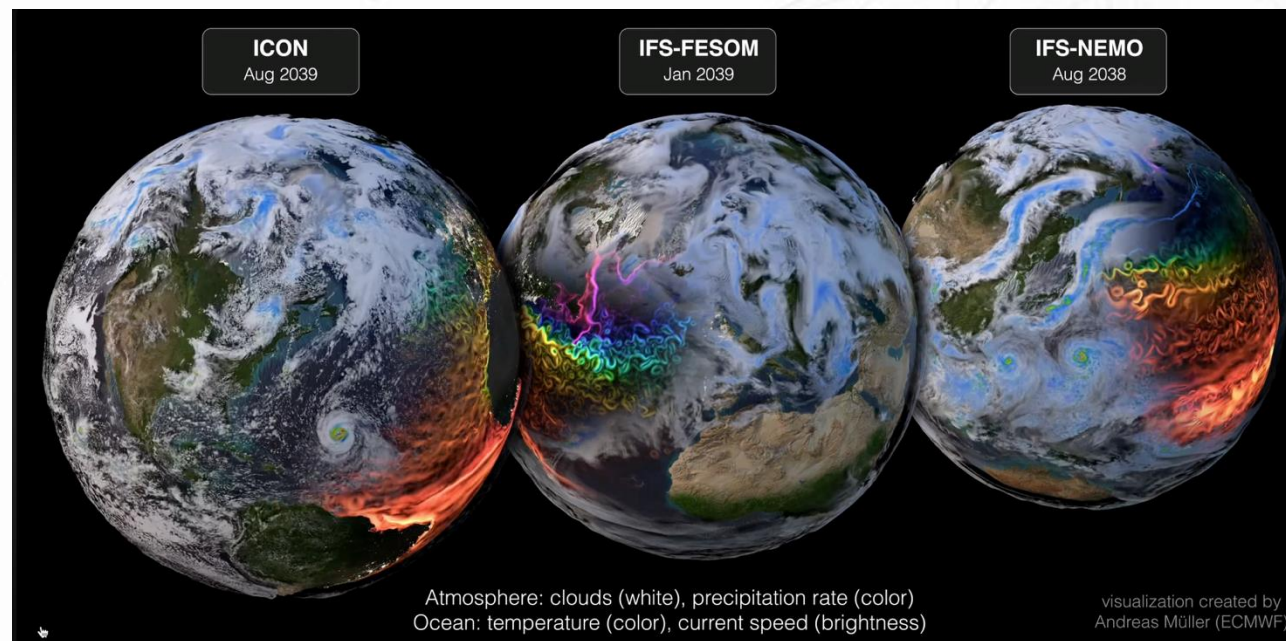


How would the European heatwave look like in a +2C world?

How much wind energy production can we expect during a windstorm?



DestinE Digital Twin Data



Central Site



LUMI



EUMETSAT Bridge



Leonardo



MareNostrum



Euro High Performance Computing (EuroHPC)



User-Generated Data

Data generated by users using their own algorithms on DestinE Data
This data is made accessible to the user community following review

User-Driven Evolutions

Data providers, Collections and functionalities will evolve over time based on DestinE user needs

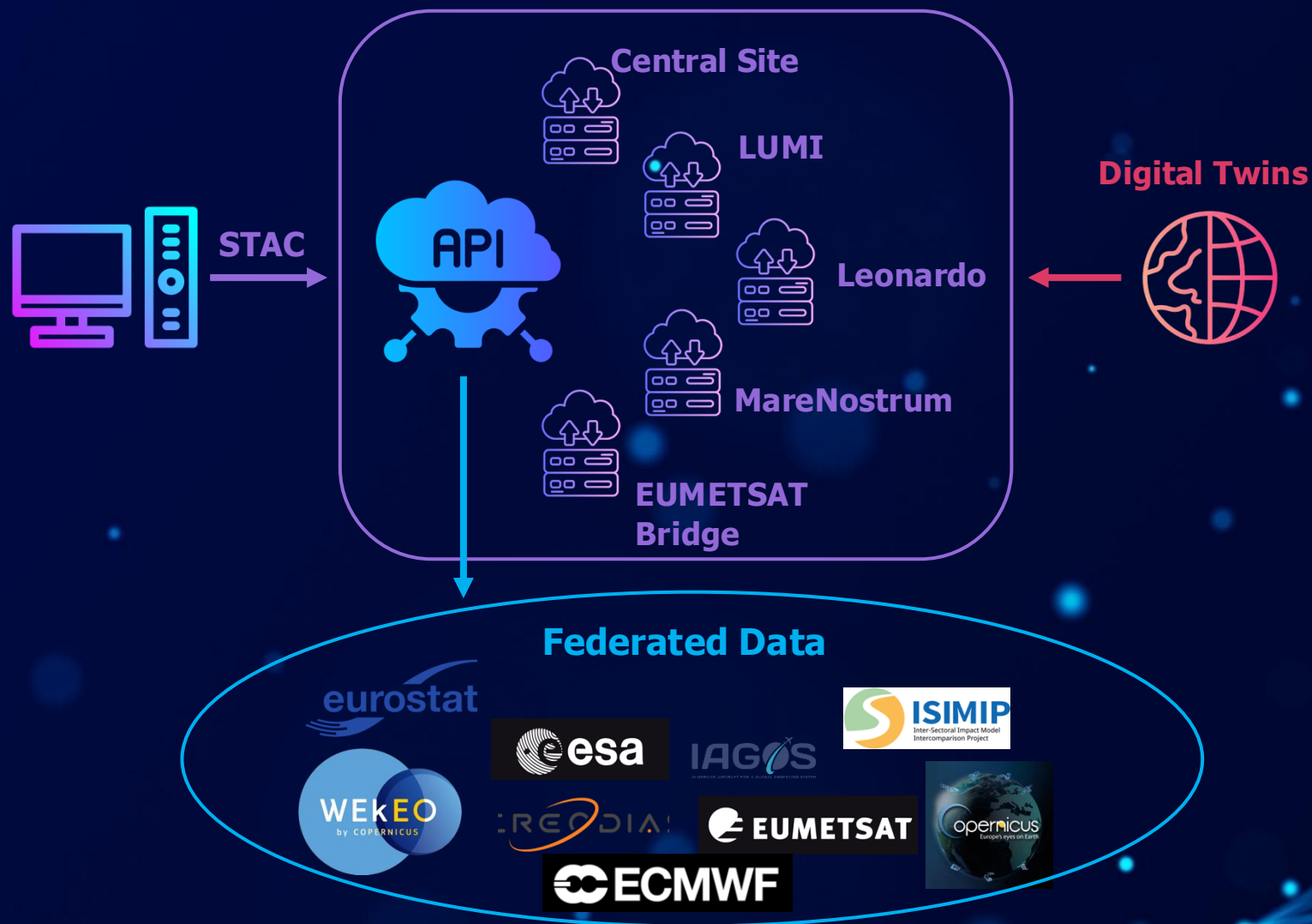




Harmonised Data Access

Offers seamless access to both DestinE Data and Federated Data via GUIs and APIs, regardless of data type and location

More specifically, it provides a common catalogue and a standardised interface, so that users can search for a dataset, define what they need and then access the data without having to know where they are physically stored or how that particular provider makes them available.



DATA LAKE EDGE SERVICES

Computing and storage capabilities provided within the DestinE Data Lake to enable users to process data close to where it is stored



DATA LAKE EDGE SERVICES

Minimises data transfer overhead and supports workflow execution for Big Data analytics, AI/ML applications and integration with Digital Twin outputs



Portfolio of Edge Services



ISLET SERVICE

INFRASTRUCTURE & TOOLS



STACK SERVICE

HOSTED APPLICATIONS



HOOK SERVICE

FUNCTIONS

Infrastructure & Tools



ISLET SERVICE

- Cloud Computing Service based on OpenStack technology
- Allows to manage and deploy virtualised workloads on a scalable and flexible architecture
- Virtual resources can be managed by users using the Horizon GUI, API or CLI
- Kubernetes can be used to deploy containerised workloads

Hosted Applications



STACK SERVICE

- Allows users to execute their algorithms and codes on the DestinE Data Lake site, close to the data of interest
- Provides solutions based on JupyterHub, Dask/Dask Gateway and Open Data Cube to facilitate processing
- Enables large-scale data processing in parallel using Dask, with results collected and saved directly to project storage

Functions



HOOK SERVICE

- Serverless execution environment for running container-based workflows on demand in the Data Lake
- Enables users to run ready-to-use functions and workflows through a simple API, without needing Kubernetes expertise
- Predefined hooks for efficient data access and processing, or create custom hooks tailored to project-specific analysis workflows



How Can I have Access
to the Data and to the
Edge Services?





Onboarding to DestinE Data Lake Edge Services

 **Destination Earth**

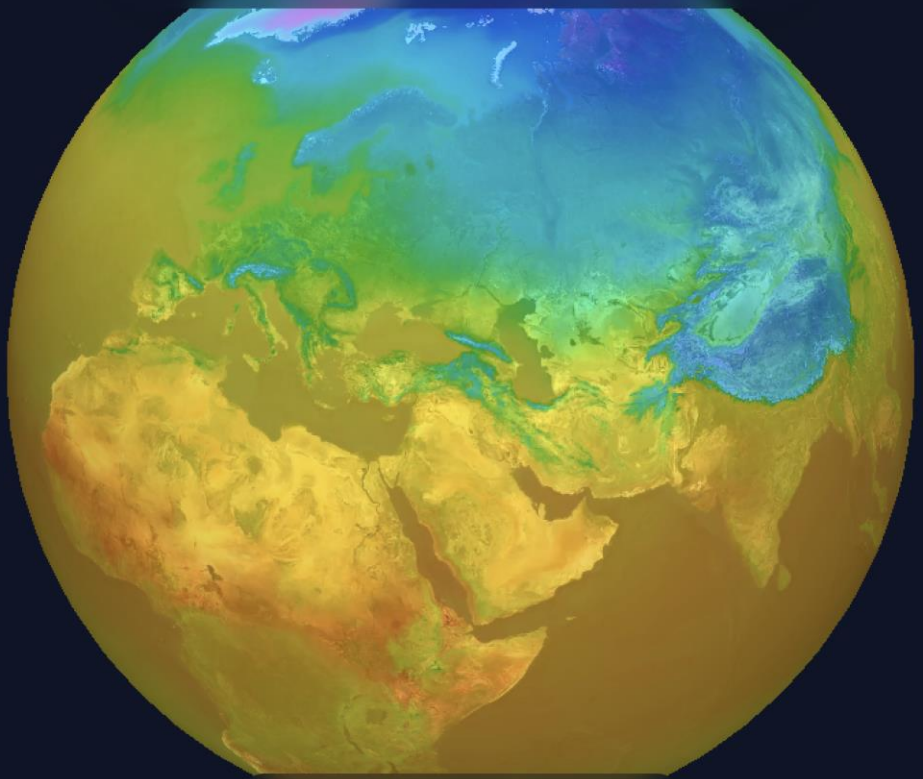
[Home](#) [Services](#) [Onboarding](#) [Updates](#) [About](#) [Resources](#)

   [Sign In](#)

Global temperature from DestinE Climate DT - GEN 2

2015  2050





Monthly 2m temperature with IFS-NEMO model (°C)

 -68 52

Powered by 



DestinE Platform

Your gateway to a sustainable future

A unique ecosystem of services harnessing the power of Destination Earth.

NEW - GEN 2

Climate DT

[Explore Climate DT](#)

10.276

Registered Users

[Register](#)

51

Services already available

[Discover](#)



Onboarding to DestinE Data Lake Edge Services

 **Destination Earth**

Home Services ▾ Onboarding Updates About Resources ▾



 joanabrt

Destination Earth User Access Upgrade

The Destination Earth User Access Policy allows users in the categories below to apply for an upgraded access to Destination Earth System, including advanced features and services.



	General public	+		Public authority	+
	Academia & research	+		Private sector	+
	3rd country public authorities	+		DestinE 3Es (ESA/ECMWF/EUMETSAT) staff	+
	DestinE 3Es (ESA/ECMWF/EUMETSAT) contractors	+		Federated ecosystem	+



DestinE Use Cases

- **A concrete user need or problem**
Something that matters to decision-makers, industry, scientists, or society — e.g. climate adaptation, extreme events, water management, energy, agriculture or marine management
- **Multiple data sources**
It benefits from combining Earth Observation, Digital Twin outputs, observations, historical datasets and/or socio-economic data, rather than relying on a single dataset
- **Large or complex data processing**
The problem is sufficiently data-intensive that DestinE's computing infrastructure and Edge Services provide a real advantage
- **Integration of modelling and observations**
Ideally, the use case connects observations with simulations or forecasts
- **A useful end product**
The output should translate the data into something usable — such as an indicator, forecast, risk assessment, scenario, dashboard or decision-support tool.

Demonstration

Onboarding to the Data Lake

- From DestinE to Data Lake
- Onboarding and My Services



FROM DATA TO INSIGHTS:

Working with DestinE Data Lake Services

PART 3

Data discovery with the HDA

Ben Loveday (Innoflair / EUMETSAT)
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[#datalake](https://twitter.com/datalake)



- The Harmonised Data Access (HDA) service is the main gateway to data within Destination Earth.

HARMONISED DATA ACCESS

Access to HDA is freely available to ALL users who have a DestinE Platform Account (a.k.a DESP) account.





Harmonised Data Access: capabilities

- The Harmonised Data Access (HDA) service is the main gateway to data within Destination Earth.
- It provides a single, unified interface (API) for discovering, searching, and accessing datasets through a single credential.



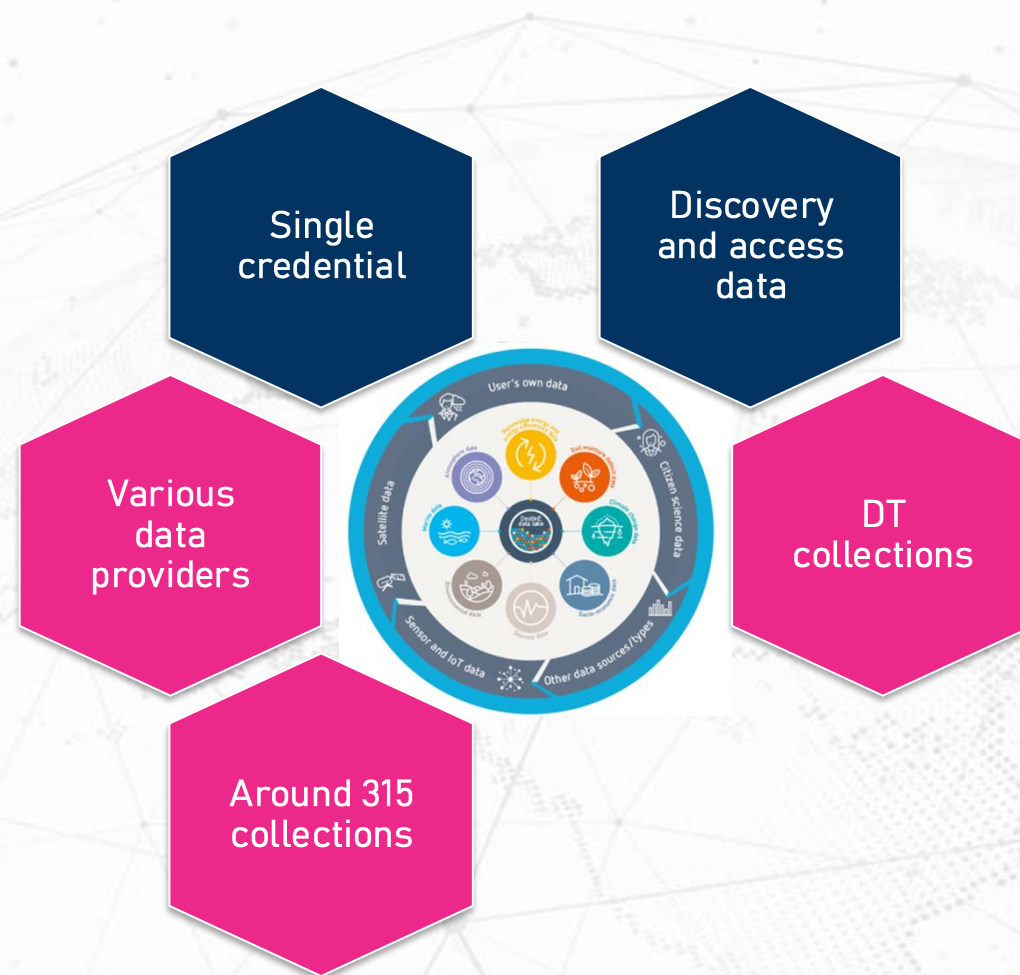


Harmonised Data Access: capabilities

- The Harmonised Data Access (HDA) service is the main gateway to data within Destination Earth.
- It provides a single, unified interface (API) for discovering, searching, and accessing datasets through a single credential.
- Uses a federation first approach to provide access to over 300 data sets from multiple federated sources, including Digital Twin outputs, user-generated data and the fresh data pool.



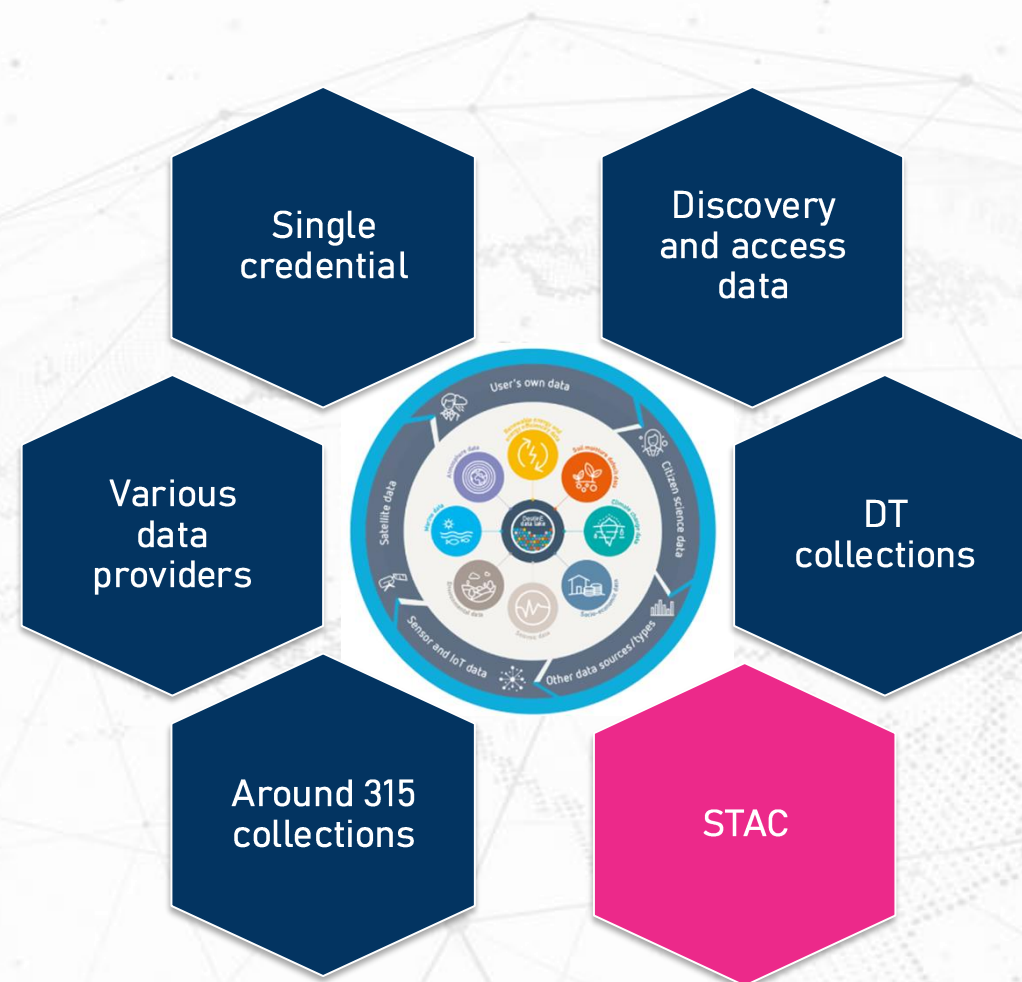
**FRESH
DATA
POOL**





Harmonised Data Access: capabilities

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- Uses a federation first approach to provide access to over 300 data sets from multiple federated sources, including Digital Twin outputs, user-generated data and the fresh data pool.
- Built around the STAC (v2) standard.





- SpatioTemporal Asset Catalog (STAC) v1.1.0
 - [STAC collection specification](#)
 - [STAC Item specification](#)
- Dataset discovery filters
 - Free-text search
 - Temporal extent
 - Area of interest
 - CQL filters
- Product search filtering capabilities
 - Temporal extent
 - Area of interest
 - Custom properties per dataset (e.g. cloud cover)
 - Specific id
 - Sort

STAC API STAC API v1.0.0. New version of STAC API for HDA API.

GET `/stac/v2` DEDL STAC capabilities discovery

GET `/stac/v2/conformance` information about specifications that this API conforms to

GET `/stac/v2/queryables` Get the JSON Schema defining the list of variable terms that can be used in CQL2 expres

GET `/stac/v2/collections` The feature collections in the dataset.

GET `/stac/v2/collections/{collectionId}` Describe the feature Collection for the given 'collectionId'.

GET `/stac/v2/collections/{collectionId}/queryables` Get the JSON Schema defining the list of variab

GET `/stac/v2/collections/{collectionId}/items` List of items available in a given collection

GET `/stac/v2/collections/{collectionId}/items/{itemId}` Fetch a single feature.

GET `/stac/v2/search` Search STAC items with simple filtering.

POST `/stac/v2/search` Search STAC items with full-featured filtering.

POST `/stac/v2/collections/{collectionId}/order` Order data from a collection. Only ECMWF-like datasets



Providing DEDL credentials



- Supported by
 - Python PyPi package [destinelab](#)
 - [EODAG](#)
- “Data Lake API key” for machine-to-machine communication
- Different levels of access:
 - No authentication required
 - Public
 - Upgraded access for DT data
 - Restricted



- Basic documentation for HDA and EODAG features: [Destination Earth Data Lake Documentation](#)
- Many notebooks with examples for different types of collections: [DestineE Data Lake Github](#) and <https://destination-earth.github.io/DestinE-DataLake-Gallery/hda/>
- Code examples at <https://github.com/destination-earth/DestinE-DataLake-Lab>



Demonstration

Data discovery with the HDA

- STAC catalogue exploration
- Filtering
- Downloading



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FROM DATA TO INSIGHTS: Working with DestinE Data Lake Services

PART 4

Data Processing with the Data Lake Edge Services

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Data Lake Edge Services: a short 101



STACK

Software as a Service
(SaaS)

Access to ready-to-use
software.

*"Give me an environment
where I can analyse the
data."*



HOOK

Function as a Service
(FaaS)

Run functions without
managing infrastructure.

*"Tell DEDL what
standard processing I
want done."*



Infrastructure as a
Service (IaaS)

Access to computing
infrastructure.

*"Give me the
infrastructure and I'll
manage what runs on it."*



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- Allows users to execute their algorithms and codes on the DestinE Data Lake site, close to the data of interest
- Provides solutions based on JupyterHub, Dask/Dask Gateway and Open Data Cube to facilitate processing
- Enables large-scale data processing in parallel using Dask, with results collected and saved directly to project storage
- Support users who want:
 - to configure ready-to-use applications and environments



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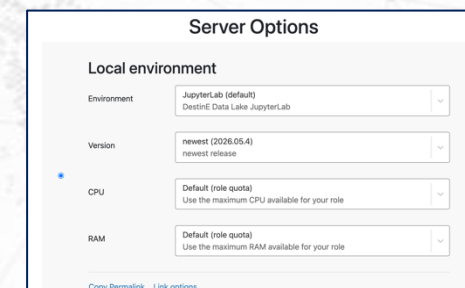
*"Give me an environment
where I can analyse the
data."*

Stack Services: ready-to-use, multitenant processing solutions, running on the Data Lake, enabling Data Processing Services.

Jupyter Hub & JupyterLab, DEDL instance of JupyterHub, acting as proxy for authenticated user to spawn JupyterLab instances

Jupyter Enterprise Gateway, enables users to launch Jupyter. kernels at the individual data sites of the distributed infrastructure

Dask Gateway, enables users to request Dask clusters on-demand to be used to distribute compute jobs across all federation sites and executing computations next to the data.





Data Lake Edge Services: STACK Jupyter



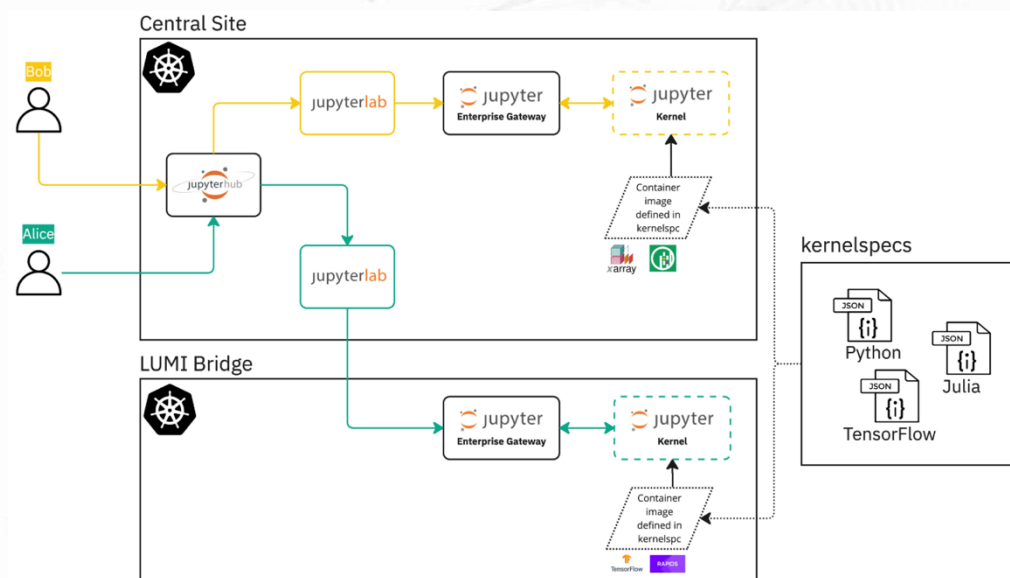
STACK

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JupyterLab: real time code environment with several extensions and plugins (git, S3, EODAG, Dask...) and a specific Python DestinE Data Lake Kernel



Server Options	
Local environment	
Environment	JupyterLab (default) DestinE Data Lake JupyterLab
Version	newest (2026.05.4) newest release
CPU	Default (role quota) Use the maximum CPU available for your role
RAM	Default (role quota) Use the maximum RAM available for your role

[Copy Permalink](#) [Link options](#)

The Jupyter Kernels available via Jupyter Enterprise Gateway can be directly accessed via a JupyterLab instance spawned in DEDL JupyterHub.



Data Lake Edge Services: STACK – DASK



STACK

Software as a Service
(SaaS)

Access to ready-to-use
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*"Give me an environment
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data."*

Stack Services: ready-to-use, multitenant processing solutions, running on the Data Lake, enabling Data Processing Services.

Remote Cluster Access: Manage and execute tasks on remote clusters directly from JupyterLab

Real-Time Dashboards: Monitor performance, task progress, and resource usage via interactive dashboards

Effortless Scaling: Dynamically balances workloads across workers for scalable distributed computing





Data Lake Edge Services: STACK – DASK



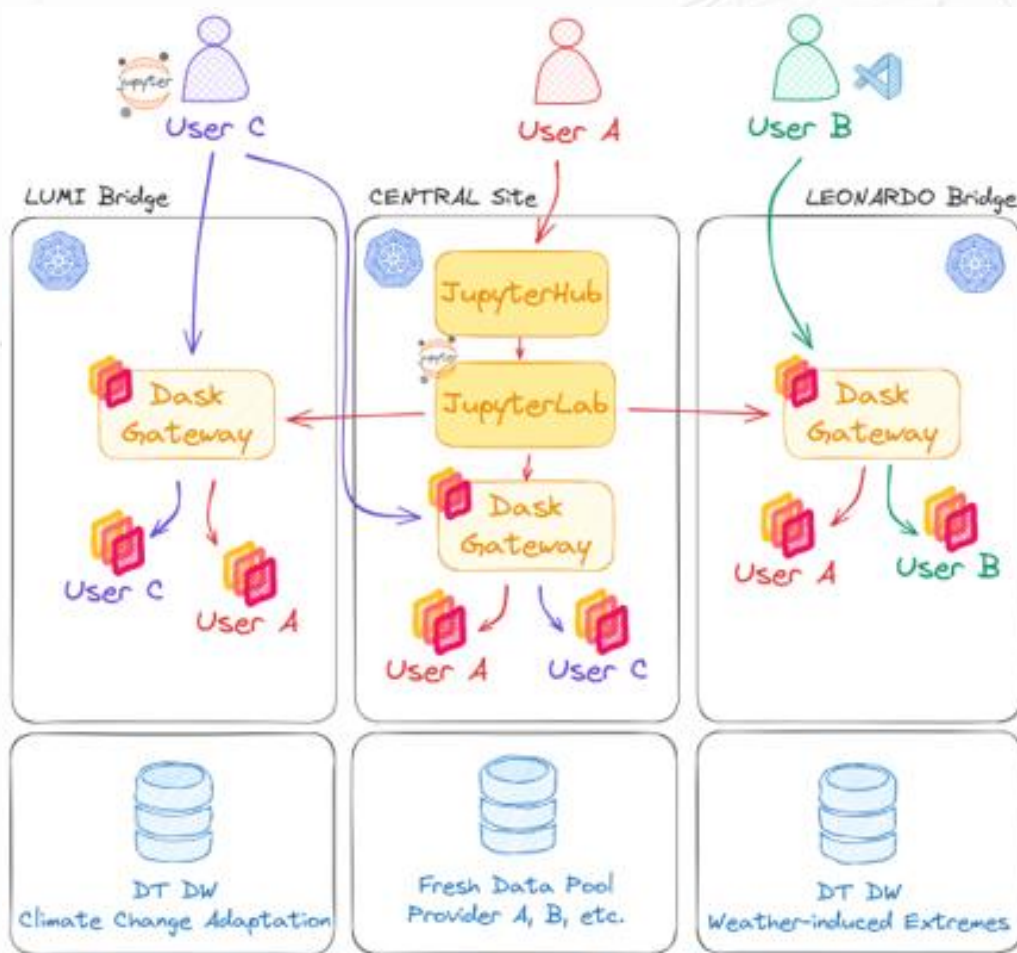
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Function as a Service (FaaS)

Run functions without
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*"Tell DEDL what
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- Serverless execution environment for running container-based workflows on demand in the Data Lake
- Enables users to run ready-to-use functions and workflows through a simple API, without needing Kubernetes expertise
- Predefined hooks for efficient data access and processing, or create custom hooks tailored to project-specific analysis workflows
- Support users who want:
 - ready-to-use building blocks for their application
 - advanced processing services



Infrastructure as a Service (IaaS)

Access to computing infrastructure.

"Give me the infrastructure and I'll manage what runs on it."

- Cloud Computing Service based on OpenStack technology
- Allows to manage and deploy virtualised workloads on a scalable and flexible architecture
- Virtual resources can be managed by users using the Horizon GUI, API or CLI
- Kubernetes can be used to deploy containerised workloads
- Support users who want:
 - to set up and manage their own development environment
 - to deploy already existing processing chains



Data Lake Edge Services: ISLET



Infrastructure as a
Service (IaaS)

Access to computing
infrastructure.

*"Give me the
infrastructure and I'll
manage what runs on it."*

- Cloud Computing Service based on OpenStack technology

Deployed across 5 data bridges.

Central (Poland) (FDP)
LUMI (Finland) – HPC / ExtremesDT
Leonardo (Italy) – HPC / ClimateDT
Marenostrum (Spain) – HPC
EUMETSAT (Germany)





Data Lake Edge Services: ISLET



Infrastructure as a
Service (IaaS)

Access to computing
infrastructure.

*"Give me the
infrastructure and I'll
manage what runs on it."*

- Cloud Computing Service based on OpenStack technology

Deployed across 5 data bridges.

Provides virtual environment based on the
OpenStack, with:

- preconfigured images
- vCPU and GPU access
- Kubernetes clusters.
- Managed through Horizon dashboard

Provides access to, and management of S3
buckets through ISLET Storage



Demonstration

Data processing with STACK

- Accessing STACK
- Setting up DASK
- Data proximate processing



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FROM DATA TO INSIGHTS:

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Services

PART 5

New tools and services: DEFAIR,
DEDL Cube & MLOPS

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DEFAIR

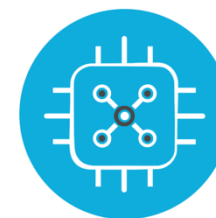
Destination Earth Framework
for preparing AI-Ready Data

- DEFAIR – Destination Earth Framework for preparing AI-Ready Data
- Python-based framework that helps defining reusable EO data preparation pipelines.
- Supports processing steps such as filtering, reprojection, collocation, alignment,...
- Enables generation of AI ready outputs in formats such as Zarr, NetCDF and GeoTIFF.
- Supports reproducible preprocessing for analytics and AI/ML workflows



DEDL CUBE

- DEDL Cube – structured access to multidimensional EO data
- Transforms diverse EO products into cloud native data cubes
- Enable efficient multidimensional access for time-series and analysis
- Built around open standards: GeoDataCube API, Open EO and Zarr/GeoZarr
- Designed to support AI/ML workflows



MLOps STUDIO

- MLOps platform – Support AI/ML lifecycle
- Managed Kubeflow solution that scaling training, pipelines and model serving.
- Provides a path from notebooks to reproducible ML workflows.
- Roadmap: Integration with Airflow and mlflow for orchestration, tracking and registry





FROM DATA TO INSIGHTS:

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--- PART 6 Conclusions & Q&A

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Takeaways



An interaction reminder...

- You can still post on slido!



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Thank you!

Questions are welcome.

You can contact us at:

users-dedl@eumetsat.int